Federated Coupled Contrastive Learning on Remote Sensing Image Classification

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***Abstract*—Classification for remote sensing (RS) images plays a crucial role in various fields such as disaster assessment and urban planning. Nevertheless, due to resource limitations and privacy issues, directly transferring data to the ground for high-performance centralized training is often impractical. Federated learning (FL) has emerged as a revolutionary distributed learning paradigm, which enables collaborative training of a model without data sharing. However, since the RS data is distributed across numerous participants, FL suffers from an inevitable challenge of data heterogeneity for RS image classification. As the local data is more unevenly distributed, this challenge becomes increasingly pronounced, leading to inconsistent local update directions. In this article, we propose RS-CCL, a novel federated coupled contrastive learning rule on RS image classification, which expands the feature space, thereby allowing models to better learn the intrinsic structure of the complex RS data effectively. Our RS-CCL mathematically formulates cross-entropy and feature similarity into a unified framework. It is then solved by coupled gradient computation in a consistent optimization direction, overcoming dimensional collapse. In this way, an enriched feature space is created, thereby facilitating the training of more capable models. Within this framework, we further introduce a dynamic fusion mask matrix, which inhibits the model from excessively drawing on interclass similarity. Empirically, we show that our RS-CCL is in line with robustness theory that a larger feature space brings more robust performance. Extensive experiments on several benchmarks validate the effectiveness and generalization ability of our method. For example, with a high data heterogeneity of $\alpha = 0.05$, our RS-CCL outperforms other baselines by up to 8.5% on the UC Merced Land Use (UC Merced) dataset with MobileViT-S. The code is available at https://github.com/cxy-xd/RS-CCL**

***Index Terms*—Federated learning (FL), remote sensing (RS), supervised contrastive learning (SCL).**

I. Introduction

With the continuous advancement of remote sensing (RS) technology, the scale of RS data has become increasingly large [1], resulting in a growing demand for effective data analysis and processing. In recent years, the evolution of deep neural networks (DNNs) has provided powerful technical support for RS data analysis, enhancing both accuracy and efficiency. Specifically, the remarkable image processing capabilities of DNNs have been successfully applied to complex RS scenarios, particularly in fields such as disaster assessment, civil infrastructure, urban planning, and environmental monitoring [2], [3], [4], [5], showcasing significant development potential.

However, the explosive growth of RS data has led to a steep rise in the training costs of deep learning models. Meanwhile, several challenges hinder the efficient training of these networks on high-performance ground equipment.

1) Resource-constrained devices such as drones struggle to store and transmit large volumes of high-resolution RS images effectively [6]. Specifically, most consumer-grade and light industrial drones only support storage capacities of a few dozen gigabytes, which can quickly become insufficient during extended missions or multi-sensor data collection.
2) The window period for communication between satellites and ground stations is restricted, preventing real-time and efficient downlink of the collected data [7]. At an altitude of 550 km, an orbiting satellite of Starlink is moving with a relative speed of 27 000 km/h relative to a point on the surface of the Earth and passes across the sky from horizon to horizon in under 5 min.
3) RS images often contain sensitive information, and directly transmitting or downlinking these data may pose privacy leakage issues [8]. For example, satellites can monitor the movements and schedule of an individual, such as showing when a car belonging to someone is parked on their driveway. Satellites can also generate images of private areas that are not visible from the ground, such as fenced-in backyards and parts of houses visible through windows or skylights.

In response to the above issues, conducting collaborative training of models in a distributed manner in space presents an attractive and feasible option. Federated learning (FL), as a subset of distributed learning, effectively addresses this challenge by enabling multiple participants to cooperatively train a global model without the need to share their individual datasets. In a typical FL framework, clients receive the global model from a central server, train it using their local data—which remains inaccessible to other participants—and

Received 12 October 2025; revised 28 December 2025; accepted 12 January 2026. Date of publication 19 January 2026; date of current version 28 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62531018, Grant 62322117, Grant 62371365, Grant U24B20136, and Grant U22B2014; and in part by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China under Grant JYB2025XDXM105. *(Corresponding author: Weiying Xie.)*

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Digital Object Identifier 10.1109/TGRS.2026.3654941

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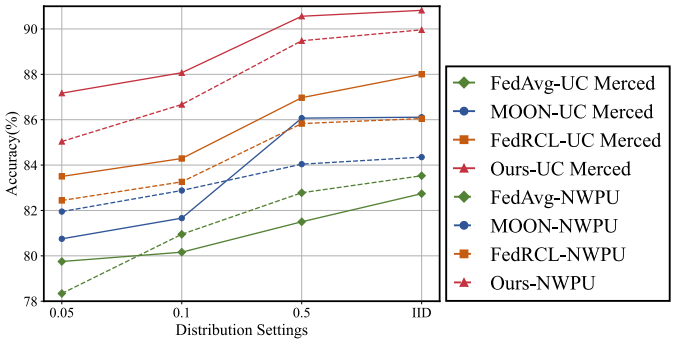


Fig. 1. Comparison of our RS-CCL against FedAvg, MOON, and FedRCL on UC Merced and NWPU datasets with the ResNet18 model. Our RS-CCL shows improvement over other methods and remains robust under high data heterogeneity ($\alpha = 0.05$).

subsequently upload the trained model back to the server. The server then aggregates these local models, employing techniques such as average aggregation or other strategies. By leveraging these fundamental characteristics, FL technologies have found extensive applications across various fields and domains [9], [10], [11].

However, a potential issue in FL is that, the RS data collected by different satellites are usually not independently and identically distributed (non-IID), a phenomenon termed as "data heterogeneity." In this scenario, collaborative model training across devices can result in inconsistent update directions due to the uneven data distribution. Moreover, as datasets across clients grow in scale and diversity, this non-IID nature hinders model convergence and generalization [12].

To address the data heterogeneity challenge, some works attempt to constrain local updates by incorporating a gradient alignment penalty or variance correction terms. FedProx [13] enhanced the stability and convergence of the global model by introducing a regularization term during the local training phase. The idea of SCAFFOLD [14] is similar to FedProx, which also regularizes the gradient. Meanwhile, some efforts focus on the aggregation scheme. FedAU [15] adaptively adjusted the weights during model aggregation phases based on an online estimation of the client's participation history. Based on theoretical analysis, Wang et al. [16] proposed FedNova, which eliminates goal inconsistency by adjusting the local update weights of clients. In addition, there are some efforts [17], [18] aimed at alleviating the data heterogeneity issue in RS tasks. While these approaches effectively mitigate the performance degradation problem due to data heterogeneity, they either focus on the local model training phase or the global aggregation phase in FL, neglecting the feature space optimization. This may lead to dimensional collapse or insufficient model characterization capability.

In this article, we propose RS-CCL, a novel federated coupled contrastive learning rule on RS image classification, which achieves coupled and consistent training directions, aiming to improve model performance in high data heterogeneity environments. The motivation behind our RS-CCL comes from the observation that a consistent learning direction facilitates feature learning, allowing the model to explore a more sufficient dimensional space. Building on this insight, we develop a mathematical formula to couple cross-entropy and feature similarity into a unified framework. Our method effectively tackles dimensional collapse by maintaining a consistent optimization direction, benefiting from cooperative gradient computation. Besides, within the unified framework, we introduce a dynamic fusion mask matrix that changes according to the model input. This matrix helps inhibit the model from excessively drawing on interclass similarity to some extent. Empirically, we experimentally demonstrate that our RS-CCL aligns with robustness theory, which suggests that a larger feature space brings about improved performance robustness. Comprehensive evaluations across two RS classification datasets and three deep learning backbones exhibit the superior performance and satisfactory generalization capability of our RS-CCL (shown in Fig. 1). For instance, with a high data heterogeneity degree of $\alpha = 0.05$, our RS-CCL outperforms other baselines by up to 2.7% in top-1 accuracy on the NWPU-RESISC45 (NWPU) RS image classification dataset.

To sum up, our contributions are as follows.

1) We propose RS-CCL, a novel FL method for RS image classification, which expands the feature space, thereby allowing the model to better extract the intrinsic structure and information of the complex RS data.
2) We couple supervised cross-entropy and supervised feature similarity into a unified loss manner, thus making both optimizations in the same direction and avoiding the dimensional collapse.
3) Extensive experiments on several benchmarks validate the superiority of our method. Notably, with a high data heterogeneity degree of $\alpha = 0.05$, our RS-CCL outperforms other baselines for comparison by up to 8.5% in top-1 accuracy on the UC Merced dataset.

The remainder of this article is structured as follows. Section II provides a review of recent advancements in FL and contrastive learning, including works that integrate FL within RS. In Section III, the foundational concepts of FL and contrastive learning are first outlined, ensuring a clear understanding of the principles that underpin our research. To this end, we reveal the dimensional collapse phenomenon in FL through a set of experiments, and propose RS-CCL based on the findings. Subsequently, Section IV experimentally validates our RS-CCL from multiple perspectives, demonstrating its effectiveness and robustness through a series of rigorous experiments. And finally, Section V concludes the article, and the future research directions are revealed.

II. Related Work

A. Federated Learning

Since FedAvg [19] first introduced the concept of FL, it has become a research hotspot. For studies on FL, most of the approaches focus on solving an important problem in this field—the problem of data heterogeneity.

Some work addresses data heterogeneity by adding constraints on local updates. Li et al. [13] proposed a kind of regularized learning method to achieve stable training

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by suppressing local updates from each client. SCAFFOLD [14] similarly accomplished this by constraining the gradient. Meanwhile, some work has turned attention to distillation-based methods to tackle the data heterogeneity problem. FedMD [20] sent the prediction scores for a portion of shared data from the clients to the server, and the average score is regarded as the global consensus of the clients. Lin et al. [21] adopted the idea of KD to fine-tune the global model after aggregation. FedGen [22] constructed a global data distribution by asking each client to upload the number of labels for each sample category, realizing the combination of FL and KD in a data-free manner.

Some FL methods focus on a more extreme type of data heterogeneity, the class imbalance problem. FedOVA [23] split the multiclass classification problem into multiple two-class classification problems and combined their results using integrated learning. Inspired by the strong generalization of the simplex equiangular tight frame (ETF), Fan et al. [24] proposed a new method called FedGELA, in which the classifier is globally fixed to the simplex ETF while locally adapting to the individual distribution.

In addition to data heterogeneity, there are several efforts [25], [26] focusing on communication-efficient FL, i.e., cost reduction in model or update transmission. Still other works [27], [28], [29] focus on personalized FL (pFL), where the server-side and client-side may have different optimization goals. Our work is orthogonal to these types of works, and can be combined with their approaches to obtain better results.

B. FL With RS

Zhang et al. [17] proposed a federated deep learning method combining prototype matching (FedPM), which shared gradient information among multiple clients instead of the original data. FedPM protects privacy while improving the performance of target extraction in high-resolution RS images. Li et al. [30] proposed a multiparty personalized collaborative learning (MPCL) framework. By decoupling the optimization of personalized models from the global model learning in each round of communication, each participant can explore personalized local models based on their own data characteristics, while a differentiated global model is provided at the task level to adapt to different RS tasks. Without sharing the original data, RS-IML [18] utilizes the intrinsic mask mechanism to enhance the feature expression ability and model generalization performance in FL. FCIL-MSN [31] is a federated class-incremental learning method combining the class-incremental strategy with the federated optimization mechanism, which effectively enhanced the model's ability in identifying new categories and generalization across satellites while also ensuring data privacy and communication efficiency. Li et al. [32] proposed FedFusion, an FL method based on a manifold-driven approach, aiming to achieve efficient fusion and collaborative modeling of multisatellite and multimodal RS data. It constructs a shared manifold space to guide each participant to perform feature alignment and knowledge transfer while preserving the local data privacy, thereby enhancing the RS image understanding capabilities across modalities and platforms. FedRSCLIP [6], as a method for RS scene classification tasks, achieved privacy protection and semantic alignment by conducting contrastive learning using image–text pairs locally among all participants, constructing a unified semantic space, and sharing model parameters rather than raw data within the federated framework. All the efforts mentioned above have demonstrated the feasibility of integrating FL into RS.

C. Contrastive Learning

Unsupervised contrastive learning, a method used to learn feature representations in deep learning, is a hot research topic in self-supervised learning. The core idea of unsupervised contrastive learning is to make positive pairs (samples augmented from the same instances) closer while pushing negative pairs (samples augmented from different instances) away from each other by learning the internal features of an image and judging them with some metrics (e.g., cosine similarity). SimCLR [33] is one of the most typical and widely used unsupervised contrastive learning methods. SimCLR proposed a new contrast loss function, InfoNCE, which can effectively distinguish between positive and negative samples, thus improving the quality of representation learning. The calculation process of InfoNCE for a positive pair (i, j) can be expressed as follows:

$$l_{i,j} = -\log \frac{\exp\left(\text{sim}\left(z_i, z_j\right)/\tau\right)}{\sum_{k=1}^{2N} \exp\left(\text{sim}\left(z_i, z_k\right)/\tau\right)} \tag{1}$$

where z_i and z_j denote the features obtained after the augmented samples i and j go through the feature encoder, respectively. $\text{sim}(x, y)$ denotes the cosine similarity between x and y, and τ denotes the temperature parameter. N denotes the number of samples for loss computation. Most of the subsequent work on contrastive learning has been carried out based on the initial InfoNCE loss function. MoCo [34] treated contrastive learning as a dictionary lookup task, while utilizing queuing and moving average encoders to construct dynamic dictionaries that can be dynamically updated during training. Instead of negative samples, BYOL [35] implemented contrastive learning by comparing two different neural networks (online network and target network). Xie et al. [36] proposed an unsupervised graph matching method, which adopted contrastive learning to optimize the matching process of graph structure, so that similar graphs are close to each other and different graphs are far away. Zhang et al. [37] proposed a time-series diversified self-contrastive learning (TDSCL) framework to capture temporal artifacts in videos through short-term and long-term self-contrastive learning, improving the generalization ability of face forgery detection.

Supervised contrastive learning (SCL) is an extension of contrastive learning, which can achieve better performance by fully utilizing label information. Khosla et al. [38] first extended unsupervised contrast learning to an SCL setting and proposed a new loss function for contrast learning, such that samples of the same classes are clustered together in the embedding space while samples of different classes are pushed apart. Jiang et al. [39] improved contrastive learning by introducing a "hardening function" that skews the distribution of negative sample selection toward samples closer to the anchor

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point. Pan et al. [40] focused on challenges in fine-grained action recognition, and proposed a novel semantic-aware late SCL approach.

Our RS-CCL is built on a local training framework of SCL, where we adopt MoCo as the fundamental architecture among several common contrastive learning frameworks for two main reasons. On the one hand, MoCo can update the negative features stably and efficiently through the dynamic dictionary query mechanism, without calculating all the characteristics of negative samples for each sample. On the other hand, MoCo maintains a dynamic pool (dictionary) of queue samples, making it possible to utilize more negative samples even in mini-batch training without relying on a large batch size for negative samples. Both of the factors are important in FL.

D. FL With Contrastive Learning

Some work attempts to integrate FL and contrastive learning. MOON [41] employed model-level contrastive learning, which corrects for local training by comparing the similarities between the global model and local models. FedProc [42] utilized prototypes as global knowledge to regulate local training on the client through global prototypical contrastive loss, ensuring consistency between local optimization and global optimization direction. FedPCL [43] similarly shared knowledge through class prototypes while learning based on the fusion of multiple pretrained models to reduce computational and communication burden. FedCDG [44] proposed an FL framework that uses contrastive learning to enhance domain generalization ability, achieving better model generalization under the premise of preserving privacy. We believe that the implementation of contrastive learning will be an important tool to tackle data heterogeneity problems in FL. By revealing the model's understanding on the intrinsic characteristics of complex RS data, we proposed RS-CCL based on the consistency of optimization directions.

III. Methodology

A. Problem Formulation

In a common FL setting, a group of N participating clients is represented as set $\mathcal{C} = \{C_1, C_2, \ldots, C_N\}$. Each client C_i owns a local dataset $\mathcal{D}_i$ consisting of n_i samples $\{(\mathbf{x}_{ij}, y_{ij})\}_{j=1}^{n_i}$, here $\mathbf{x}_{ij}$ denotes the jth sample on client C_i, and y_{ij} denotes the corresponding label.

We denote a global neural network parameterized by the weight vector $\mathbf{w}$ as $f_{\mathbf{w}}$. The optimization objective of $f_{\mathbf{w}}$ on dataset $\mathcal{D}$ ($\mathcal{D} = \bigcup_{i=1}^{N} \mathcal{D}_i$) is to minimize the empirical risk

$$F_i(\mathbf{w}_i, \mathcal{D}_i) = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathcal{L}\left(f_{\mathbf{w}_i}(\mathbf{x}_{ij}), y_{ij}\right) \tag{2}$$

$$F(\mathbf{w}, \mathcal{D}) = \sum_{i=1}^{N} \frac{n^i}{n} F_i(\mathbf{w}_i, \mathcal{D}_i) \tag{3}$$

where $f_{\mathbf{w}}(\mathbf{x}_{ij}) \in \mathbb{R}^K$ represents the predicted classification probabilities of $\mathbf{x}_{ij}$, and K is the number of classes. In the local training phase of FL, the cross-entropy loss function is usually employed to train the local model, i.e.,

$$\mathcal{L}_{\mathrm{CE}}\left(f_{\mathbf{w}_i}(\mathbf{x}_{ij}), y_{ij}\right) = \sum_{i=1}^{K} f_{\mathbf{w}_i}(\mathbf{x}_{ij}) \log\left(h(y_{ij})\right) \tag{4}$$

where $h(y_{ij})$ denotes the label y_{ij} converted to one-hot form. Generally, in the tth communication round of FL, the global model is updated according to the following equations:

$$\mathbf{w}_i^{(t+1)} = \mathbf{w}_i^{(t)} - \eta_t \nabla F_i\left(\mathbf{w}_i^{(t)}, \mathcal{D}_i\right) \tag{5}$$

$$\mathbf{w}^{(t+1)} = \frac{1}{N \times P} \sum_{i \in \mathcal{S}_t} \mathbf{w}_i^{(t+1)} \tag{6}$$

where $\mathbf{w}_i^{(t)}$ is the local model weight of client C_i in the tth communication round, η_t represents the learning rate in the tth round, $\mathbf{w}^{(t+1)}$ is the global model weight in the $(t+1)$th round, P denotes the proportion of clients selected in each round, and $\mathcal{S}_t$ denotes the subset of clients participating in the tth round.

B. SCL in FL

For the standard cross-entropy loss function, the gradient with respect to the weight corresponding to the ith class (i.e., the ith column of the weight matrix in the classification layer) is computed as follows:

$$\begin{aligned} \frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial \mathbf{w}_{,i}^{\mathrm{linear}}} &= \frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial o_i} \cdot \frac{\partial o_i}{\partial \mathbf{w}_{,i}^{\mathrm{linear}}} \\ &= \left(\widehat{y}_i - y_j\right) \mathbf{x} \\ &= \begin{cases} \left(\widehat{y}_i - 1\right) \mathbf{x}, & \text{if } j = i \\ \widehat{y}_i \mathbf{x}, & \text{if } j \neq i \end{cases} \end{aligned} \tag{7}$$

where $\mathbf{w}_{,i}^{\mathrm{linear}}$ refers to the ith column of the classification layer weight matrix, o_i is the logits output of the ith class, $\mathbf{x}$ is the input features of the classification layer, and $\widehat{y}_i = \mathrm{softmax}(o_i)$ is the predicted probability for the ith class.

Based on the above gradient calculation results, we have the following corollary and its corresponding proof revealing the limitations of cross-entropy-based local training under non-IID settings in FL.

Corollary 1: For a classification task with K classes ($K > 2$), assume the model outputs a uniform predicted probability across all classes, i.e., $\widehat{y}_1 = \widehat{y}_2 = \cdots = \widehat{y}_K = 1/K$. For a client with N samples under non-IID data distribution (maximum class with N_1 samples, minimum class with N_2 samples), the ratio between the sums of the $L-1$ norms for the gradients received by the weight vectors in the classification layer corresponding to the two classes is approximately $((K-2)N_1 + N)/((K-2)N_2 + N)$.

Proof: Recall that

$$\frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial \mathbf{w}_{,i}^{\mathrm{linear}}} == \begin{cases} \left(\widehat{y}_i - 1\right) \mathbf{x}, & \text{if } j = i \\ \widehat{y}_i \mathbf{x}, & \text{if } j \neq i. \end{cases} \tag{8}$$

For maximum class k_1 with N_1 samples, we have

$$g_1 = \sum_{i=1}^{N_1} \left| \left(\widehat{y}_{k_1} - 1\right) \mathbf{x}_i \right| + \sum_{j=1}^{N-N_1} \left| \widehat{y}_{k_1} \mathbf{x}_j \right|$$

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$$= \sum_{i=1}^{N_1} \left| \left(\frac{1}{K} - 1 \right) \mathbf{x}_i \right| + \sum_{j=1}^{N-N_1} \left| \frac{1}{K} \mathbf{x}_j \right|$$
$$= \sum_{i=1}^{N_1} \left(1 - \frac{1}{K} \right) |\mathbf{x}_i| + \sum_{j=1}^{N-N_1} \frac{1}{K} |\mathbf{x}_j|. \tag{9}$$

Similarly, for minimum class k_2 with N_2 samples, we have

$$g_2 = \sum_{i=1}^{N_2} \left(1 - \frac{1}{K} \right) |\mathbf{x}_i| + \sum_{j=1}^{N-N_2} \frac{1}{K} |\mathbf{x}_j|. \tag{10}$$

Since $\mathbf{x}$ is computed and processed through multiple layers of normalization of the model (usually in the BatchNorm layer), it can be assumed without loss of generality that, for $|\mathbf{x}| = \sum_{i=1}^{D} |x_i|$ (D is the dimension of vector $\mathbf{x}$), $|\mathbf{x}_i| = \cdots = |\mathbf{x}_j|$ for $i = 1, 2, \ldots, N_1$ and $j = 1, 2, \ldots, (N - N_1)$ holds. In this case, we have

$$\begin{aligned} \frac{g_1}{g_2} &= \frac{\sum_{i=1}^{N_1} \left(1 - \frac{1}{K}\right) |\mathbf{x}_i| + \sum_{j=1}^{N-N_1} \frac{1}{K} |\mathbf{x}_j|}{\sum_{i=1}^{N_2} \left(1 - \frac{1}{K}\right) |\mathbf{x}_i| + \sum_{j=1}^{N-N_2} \frac{1}{K} |\mathbf{x}_j|} \\ &= \frac{\sum_{i=1}^{N_1} \left(1 - \frac{1}{K}\right) + \sum_{j=1}^{N-N_1} \frac{1}{K}}{\sum_{i=1}^{N_2} \left(1 - \frac{1}{K}\right) + \sum_{j=1}^{N-N_2} \frac{1}{K}} \\ &= \frac{N_1 \left(1 - \frac{1}{K}\right) + (N - N_1) \frac{1}{K}}{N_2 \left(1 - \frac{1}{K}\right) + (N - N_2) \frac{1}{K}} \\ &= \frac{\frac{N_1 K - N_1 + N - N_1}{K}}{\frac{N_2 K - N_2 + N - N_2}{K}} = \frac{(K-2) N_1 + N}{(K-2) N_2 + N}. \end{aligned} \tag{11}$$

This completes the proof.

From the corollary, it can be observed that the magnitude of updates produced by the cross-entropy loss with respect to a particular class correlates with the proportion of samples belonging to that class in the dataset. This explains the poor performance of cross-entropy-based local training in FL under data-heterogeneous (non-IID) settings.

In this case, we attempt to introduce SCL [38] into the local training phase of FL. Our approach builds upon a supervised variant of the momentum contrast (MoCo) framework. We integrate SCL into the MoCo framework, and construct a simple supervised MoCo loss function (denoted as $\mathcal{L}_{\text{SSMoCo}}$) with the following equation:

$$\mathcal{L}_{\text{SSMoCo}} = -\frac{1}{|P(x)|} \sum_{p \in P(x)} \log \frac{\exp\left(\text{sim}\left(\mathbf{q}_x, \mathbf{z}_p\right)/\tau\right)}{\sum_{a \in \mathcal{A}(x)} \exp\left(\text{sim}\left(\mathbf{q}_x, \mathbf{z}_a\right)/\tau\right)} \tag{12}$$

where $P(x)$ represents the set of positive samples belonging to the same class as sample x, and $\mathcal{A}(x)$ denotes the set of all contrast samples for sample x (for MoCo, usually refers to the union of the query batch, key batch, and the queue). $\mathbf{q}_x$ refers to the feature of the query image (augmented from x), $\mathbf{z}_p$ refers to the feature of sample p. $\text{sim}(a, b)$ represents a similarity measurement method (e.g., cosine similarity), and τ is the temperature hyperparameter.

We similarly show by a corollary that the gradient computation for SCL is not affected by the sample proportion under non-IID.

Corollary 2: For a class with $|P(i)|$ samples, the sum of the gradients' L1 norm by $\mathcal{L}_{\text{SSMoCo}}$ for the features $\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_{|P(i)|}$ is as follows:

$$\sum_{i=1}^{M} \left\| \frac{\partial \mathcal{L}_{\text{SSMoCo}}}{\partial \mathbf{q}_i} \right\| = \frac{1}{\tau} \left\| \overline{q} - \sum_{a \in \mathcal{A}(x)} \frac{\exp\left(\overline{q} \cdot \mathbf{z}_a / \tau\right)}{\sum_{a \in \mathcal{A}(x)} \exp\left(\left(\overline{q} \cdot \mathbf{z}_a\right)/\tau\right)} \cdot \mathbf{z}_a \right\| \tag{13}$$

where $\overline{q} = \sum_{i=1}^{|P(i)|} \mathbf{q}_i$ is the center of the features for the class.

Proof: Recall that

$$\mathcal{L}_{\text{SSMoCo}} = -\frac{1}{|P(x)|} \sum_{p \in P(x)} \log \frac{\exp\left(\mathbf{q}_x \cdot \mathbf{z}_p / \tau\right)}{\sum_{a \in \mathcal{A}(x)} \exp\left(\mathbf{q}_x \cdot \mathbf{z}_a / \tau\right)}. \tag{14}$$

For simplicity, we denote $(\mathbf{q}_x \cdot \mathbf{z}_p)$ by $s_{x,p}$, and $(\mathbf{q}_x \cdot \mathbf{z}_a)$ by $s_{x,a}$. First, calculate the gradient of $\mathcal{L}_{\text{SSMoCo}}$ with respect to $\mathbf{q}_x$

$$\frac{\partial \mathcal{L}_{\text{SSMoCo}}}{\partial \mathbf{q}_x} = -\frac{1}{|P(x)|} \sum_{p \in P(x)} \left[\frac{\partial \left(\frac{s_{x,p}}{\tau}\right)}{\partial \mathbf{q}_x} - \frac{\partial}{\partial \mathbf{q}_x} \log \sum_{a \in \mathcal{A}(x)} \exp\left(\frac{s_{x,a}}{\tau}\right) \right]. \tag{15}$$

Then, the two partial derivatives are computed independently

$$\frac{\partial \left(\frac{s_{x,p}}{\tau}\right)}{\partial \mathbf{q}_x} = \frac{1}{\tau} \cdot \frac{\partial \left(\mathbf{q}_x \cdot \mathbf{z}_p\right)}{\partial \mathbf{q}_x} = \frac{1}{\tau} \mathbf{z}_p \tag{16}$$

$$\begin{aligned} \frac{\partial}{\partial \mathbf{q}_x} \log \sum_{a \in \mathcal{A}(x)} \exp\left(\frac{s_{x,a}}{\tau}\right) &= \frac{\frac{\partial}{\partial \mathbf{q}_x} \sum_{a \in \mathcal{A}(x)} \exp\left(\frac{s_{x,a}}{\tau}\right)}{\sum_{a \in \mathcal{A}(x)} \exp\left(\frac{s_{x,a}}{\tau}\right)} \\ &= \frac{\sum_{a \in \mathcal{A}(x)} \frac{\partial \exp\left(s_{x,a}/\tau\right)}{\partial \mathbf{q}_x}}{\sum_{a \in \mathcal{A}(x)} \exp\left(s_{x,a}/\tau\right)} \\ &= \frac{\sum_{a \in \mathcal{A}(x)} \exp\left(s_{x,a}/\tau\right) \cdot \frac{\partial \left(\frac{s_{x,a}}{\tau}\right)}{\partial \mathbf{q}_x}}{\sum_{a \in \mathcal{A}(x)} \exp\left(s_{x,a}/\tau\right)} \\ &= \frac{1}{\tau} \cdot \frac{\sum_{a \in \mathcal{A}(x)} \exp\left(s_{x,a}/\tau\right) \mathbf{z}_a}{\sum_{a \in \mathcal{A}(x)} \exp\left(s_{x,a}/\tau\right)}. \end{aligned} \tag{17}$$

Combining (16) and (17), we have

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{SSMoCo}}}{\partial \mathbf{q}_x} &= -\frac{1}{\tau |P(x)|} \sum_{p \in P(x)} \left[\mathbf{z}_p - \sum_{a \in \mathcal{A}(x)} p_a \mathbf{z}_a \right] \\ &= -\frac{1}{\tau |P(x)|} \left[\sum_{p \in P(x)} \mathbf{z}_p - \sum_{a \in \mathcal{A}(x)} p_a \mathbf{z}_a \right] \end{aligned} \tag{18}$$

where p_a denotes the normalized probability distribution

$$p_a = \frac{\exp\left(s_{x,a}/\tau\right)}{\sum_{a \in \mathcal{A}(x)} \exp\left(s_{x,a}/\tau\right)}. \tag{19}$$

For $\mathbf{q}_x$, the absolute value of its gradient is expressed as

$$\left\| \frac{\partial \mathcal{L}_{\text{SSMoCo}}}{\partial \mathbf{q}_x} \right\| = \frac{1}{\tau |P(x)|} \left\| \sum_{p \in P(x)} \mathbf{z}_p - \sum_{a \in \mathcal{A}(x)} p_a \mathbf{z}_a \right\|. \tag{20}$$

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Then, for a class with a total of $|P(i)|$ samples, we sum the absolute values of the gradient in the loss function with respect to $\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_{|P(i)|}$

$$\sum_{i=1}^{|P(i)|} \left\| \frac{\partial \mathcal{L}_{\text{SSMoCo}}}{\partial \mathbf{q}_i} \right\| = \frac{1}{\tau |P(i)|} \sum_{i=1}^{|P(i)|} \left\| \sum_{p \in P(i)} \mathbf{z}_p - \sum_{a \in \mathcal{A}(i)} p_a \mathbf{z}_a \right\|. \quad (21)$$

Let $\bar{q} = \sum_{i=1}^{|P(i)|} \mathbf{q}_i$. It is obvious that $\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_{|P(i)|}$ share the same $\bar{q}$ and $\sum_{a \in \mathcal{A}(i)} p_a \mathbf{z}_a$, therefore we have

$$\begin{aligned} &\sum_{i=1}^{|P(i)|} \left\| \frac{\partial \mathcal{L}_{\text{SSMoCo}}}{\partial \mathbf{q}_i} \right\| \\ &\quad = \frac{1}{\tau} \left\| \bar{q} - \sum_{a \in \mathcal{A}(i)} p_a \mathbf{z}_a \right\| \\ &\quad = \frac{1}{\tau} \left\| \bar{q} - \sum_{a \in \mathcal{A}(x)} \frac{\exp\left(\bar{q} \cdot \mathbf{z}_a / \tau\right)}{\sum_{a \in \mathcal{A}(x)} \exp\left((\bar{q} \cdot \mathbf{z}_a) / \tau\right)} \cdot \mathbf{z}_a \right\|. \end{aligned} \quad (22)$$

This completes the proof.

It is intuitive from Corollary 2 that the gradient magnitude for samples within a given class is not sensitive to class sample proportions, but depends primarily on feature-level similarity. Therefore, it can be argued that integrating SCL into FL mitigates the bias induced by data heterogeneity.

C. Analysis of $\mathcal{L}_{SSMoCo}$

In this section, we make a simple attempt to analyze the components of $\mathcal{L}_{\text{SSMoCo}}$ in detail. Specifically, $\mathcal{L}_{\text{SSMoCo}}$ can be understood as a combination of the self-supervised loss for each sample (represented by $\mathcal{L}_{\text{UCL}}$) and the similarity loss between samples with the same label (represented by $\mathcal{L}_{\text{ICCL}}$). The formulas for both are as follows:

$$\mathcal{L}_{\text{UCL}} = -\log \frac{\exp\left(\text{sim}\left(\mathbf{q}_x, \mathbf{k}^+\right) / \tau\right)}{\sum_{a \in \mathcal{A}(x)} \exp\left(\text{sim}\left(\mathbf{q}_x, \mathbf{k}_a\right) / \tau\right)} \quad (23)$$

$$\mathcal{L}_{\text{ICCL}} = -\sum_{c \in P(x) \setminus \{\mathbf{k}^+\}} \frac{\log \frac{\exp(\text{sim}(\mathbf{q}_x, \mathbf{z}_c)/\tau)}{\sum_{a \in \mathcal{A}(x)} \exp(\text{sim}(\mathbf{q}_x, \mathbf{z}_a)/\tau)}}{|P(x) - 1|}. \quad (24)$$

Next, we analyze of the components of $\mathcal{L}_{\text{SSMoCo}}$ in FL. Specifically, we conduct experiments on UC Merced [45] dataset, using FedAvg as the baseline with 24% of 42 clients participating in each communication round. To simulate data heterogeneity, we consider three settings: independent identically distributed (IID) and two Dirichlet distributions ($\alpha = 0.5$ and $\alpha = 0.05$). A smaller α corresponds to a more heterogeneous data distribution.

We evaluate the decoupled components in two ways. First, we compare the classification accuracy of the baseline with that of the baseline augmented by each component under the three data distributions. Second, we examine how well the learned features utilize the dimensional space under different settings, which is also adopted in prior works [46], [47]. We employ two metrics: the cumulative energy of singular values and the effective rank.

Definition 1 (Cumulative Energy of Singular Values): For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, let $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$ be its singular values from SVD, where $r = \text{rank}(A)$. The squared Frobenius norm $\|\mathbf{A}\|_F^2 = \sum_{i=1}^{r} \sigma_i^2$ represents the total energy of $\mathbf{A}$. The cumulative energy captured by the first k singular values is defined as

$$E_k = \sum_{i=1}^{k} \sigma_i^2 \, (1 \leq k \leq r). \quad (25)$$

To determine how many singular values are needed to retain a specific fraction δ of the total energy, we define

$$k_\delta = \min \left\{ \frac{\sum_{i=1}^{k} \sigma_i^2}{\|\mathbf{A}\|_F^2} \, | k_\delta \in \mathbb{N} \right\}. \quad (26)$$

This measure, widely used in dimensionality reduction and low-rank approximation, indicates the compactness of the feature representation.

Definition 2 (Effective Rank): For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$ ($r = \text{rank}(\mathbf{A})$) obtained from SVD, we first normalize them into a probability distribution

$$p_j = \frac{\sigma_j}{\sum_{i=1}^{r} |\sigma_i|}. \quad (27)$$

The effective rank is defined as the exponential of the Shannon entropy of this distribution

$$\text{erank}(A) = \exp\{H(p_1, p_2, \ldots, p_r)\} \quad (28)$$

$$H(p_1, p_2, \ldots, p_r) = -\sum_{i=1}^{r} p_i \log p_i. \quad (29)$$

For the experiments on cumulative energy and effective rank, we randomly sample 50 images per class from the UC Merced test set, and perform SVD on the flattened features in the last convolutional layer of ResNet18. We evaluate both metrics at several key nodes in FL ($r \in \{100, 200, 500, 1000\}$).

Recall that $\mathcal{L}_{\text{SSMoCo}}$ [see (12)] can be conceptually decoupled into an unsupervised contrastive loss $\mathcal{L}_{\text{UCL}}$ [see (23)] and an intraclass contrastive loss $\mathcal{L}_{\text{ICCL}}$ [see (24)]. While both components aim to improve representation learning, they operate under different probabilistic normalizations: $\mathcal{L}_{\text{UCL}}$ contrasts each instance against all others in a batch, whereas $\mathcal{L}_{\text{ICCL}}$ explicitly pulls together all features belonging to the same class. A direct combination $\mathcal{L}_{\text{UCL}} + \mathcal{L}_{\text{ICCL}}$ therefore optimizes two objectives in separate probability spaces, leading to inconsistent gradient directions. This inconsistency becomes especially pronounced in FL, where local data is biased, causing the model to receive conflicting signals about whether to prioritize instance discrimination or intraclass compactness.

Empirically, we observe that while adding either $\mathcal{L}_{\text{UCL}}$ or $\mathcal{L}_{\text{ICCL}}$ to the cross-entropy loss improves classification accuracy under non-IID settings (see Fig. 2), it also leads to a reduction in the effective dimensionality of the learned feature space, as measured by cumulative singular value energy and effective rank (see Fig. 3). This indicates a tendency toward dimensional collapse. The collapse is more severe for $\mathcal{L}_{\text{ICCL}}$ than for $\mathcal{L}_{\text{UCL}}$, and the combination $\mathcal{L}_{\text{UCL}} + \mathcal{L}_{\text{ICCL}}$ performs similar to $\mathcal{L}_{\text{ICCL}}$ alone in this regard.

In contrast, $\mathcal{L}_{\text{SSMoCo}}$, which couples instance-level and class-level positives within a single, unified softmax normalization, achieves comparable accuracy to $\mathcal{L}_{\text{UCL}} + \mathcal{L}_{\text{ICCL}}$

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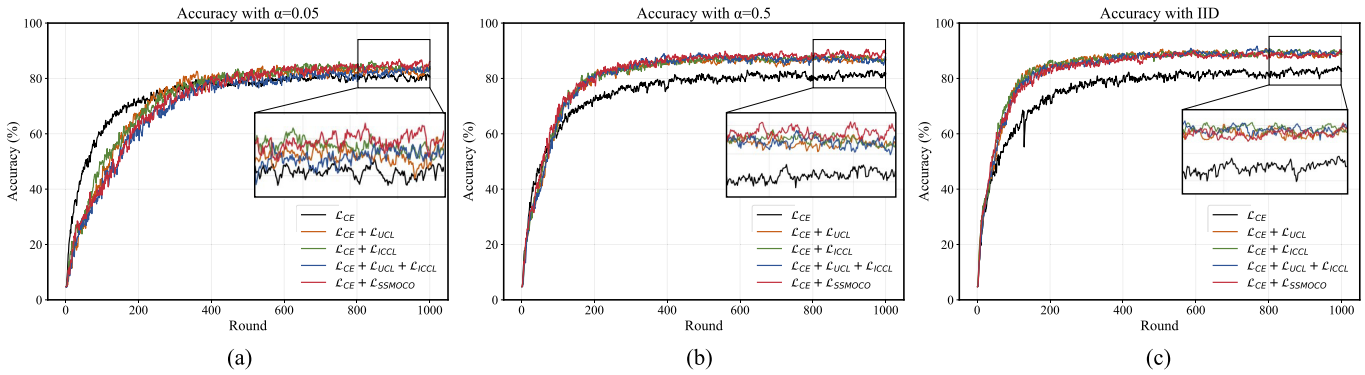


Fig. 2. Accuracy curves of implementing different components $\mathcal{L}_{\text{SSMoCo}}$ with ResNet18 on UC Merced dataset under different data distributions. (a) $\alpha = 0.05$, (b) $\alpha = 0.05$, and (c) IID.

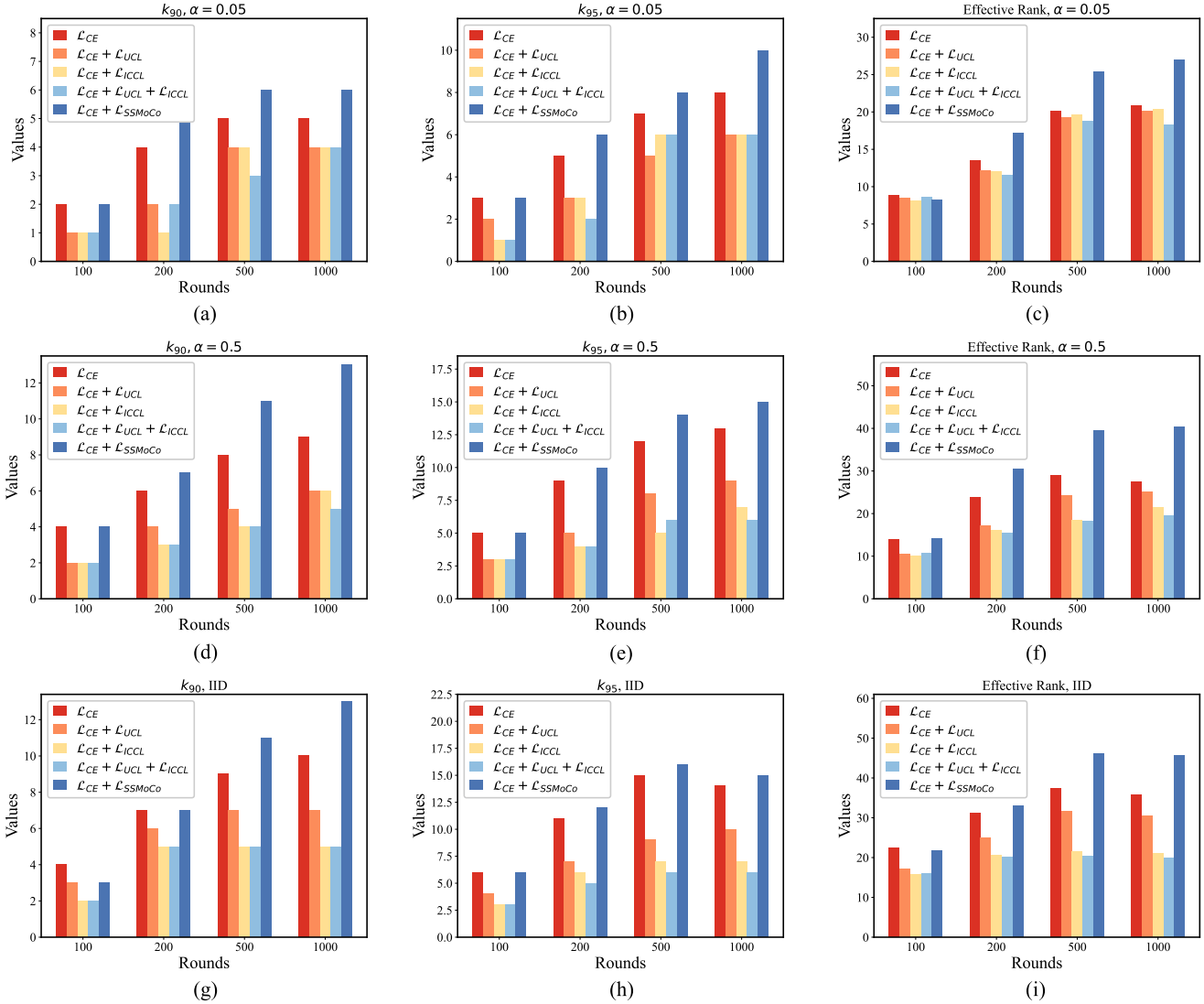


Fig. 3. Analysis of implementing different components of $\mathcal{L}_{\text{SSMoCo}}$ with ResNet18 on UC Merced dataset. Three different data distributions are applied to the experiments: Dirichlet distribution with (a)–(c) $\alpha = 0.05$ and (d)–(f) $\alpha = 0.5$, and (g)–(i) IID data distribution. $k_{0.9}$, $k_{0.95}$, and effective rank of the 100th, 200th, 500th, and 1000th communication rounds are counted.

but demonstrates superior preservation of the feature space dimensionality (see Fig. 3). The coupled formulation ensures that gradients, for instance, discrimination and intraclass alignment, are computed consistently against a shared set of negatives, preventing either objective from dominantly and independently shaping the embedding geometry. This results

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in a more balanced utilization of the feature dimensions and mitigates collapse.

The relationship to non-IID data in FL is critical. Under data heterogeneity, local clients possess skewed class distributions. Independently normalized losses like $\mathcal{L}_{\mathrm{UCL}}$ and $\mathcal{L}_{\mathrm{ICCL}}$ can over-emphasize the local data structure, leading to aggressive intraclass contraction or biased instance separation. When aggregated, these locally biased updates exacerbate global feature collapse. The unified normalization of $\mathcal{L}_{\mathrm{SSMoCo}}$ provides a more stable and consistent learning signal across clients, reducing the distorting effect of local data bias and enabling the global model to maintain a richer, more generalizable feature representation even under high heterogeneity.

In summary, the analysis reveals that a coupled contrastive formulation is key to aligning optimization directions and preserving feature dimensionality in FL, which directly motivates our proposed coupled contrastive loss in Section III-D.

D. Coupled Contrastive Loss

We attempt to couple the cross-entropy loss function with the SCL loss, and reduce the impact of interclass sample similarity on model learning. In this case, we propose a coupled supervised contrastive loss for FL. The overall calculation process of $\mathcal{L}_{\mathrm{CCL}}$ is formulated as follows:

$$\mathcal{L}_{\mathrm{CCL}} = -\frac{1}{2N}\sum_{i=1}^{2N}\left(\frac{\left[\widetilde{\mathbf{M}}\bigodot\left(\mathbf{S}-\mathrm{LSE}\left(\mathbf{S}\bigodot\widetilde{\mathbf{M}}\right)\right)\right]\cdot\mathbf{1}_F}{\left(1+\gamma\left|P\left(j\right)\right|\right)\cdot\mathbf{1}_F}\right) \tag{30}$$

where N represents the batch size, $\mathbf{1}_F$ is an all-ones vector of dimension $F \times 1$, $F = 2N + K + Q$ represents the number of features under MoCo (key batch, query batch, and queue), K represents the number of classes, and Q represents the queue size. LSE$(\cdot)$ refers to the LogSumExp operation, and $\mathbf{S}$ refers to the supervised contrastive coupled matrix, denoted as

$$\mathbf{S} = \left[\mathbf{L}\left|\mathrm{sim}\left(\mathbf{Q},\mathbf{Q}^T\right)/\tau\right.\right] \in \mathbb{R}^{2N\times(K+2N+Q)} \tag{31}$$

where $\mathbf{L}$ refers to the supervised logits matrix, and $\mathbf{Q}$ refers to the feature matrix. In general, the operation of $\mathbf{S} - \mathrm{LSE}(\mathbf{S}\bigodot\widetilde{\mathbf{M}}_i)$ is logically identical to the log-softmax operation in the cross-entropy loss, except that the logits and masked feature similarity are coupled to calculate the normalized probability distribution. $\widetilde{\mathbf{M}}$ is a dynamic fusion mask matrix, which dynamically changes as the input samples and queues change, through

$$\widetilde{\mathbf{M}} = \left[\mathbf{G}\left|\gamma\cdot\mathbf{E}\right.\right] \in \mathbb{R}^{2N\times(K+2N+Q)} \tag{32}$$

where $\mathbf{G}$ represents the ground-truth label for the inputs, and $\mathbf{E}$ represents the equality matrix, where indexes belonging to the same category are 1, and indexes in different categories are 0. γ is a hyperparameter used for dynamic mask fusion, and is set to a small number (in our experiments, 0.05) for reducing the impact of interclass feature similarity.

The overall flow of our proposed RS-CCL is shown in Algorithm 1. Our method avoids problems such as dimension collapse and model underfitting by coupling the computation of entropy information with feature-related information. This is well explained by the experimental results shown in Section IV.

Algorithm 1 Overall Procedure of RS-CCL

Input: Learning rate η; Number of clients N; Number of total communication rounds R; Local training epochs E; Proportion of clients selected in each round P; Local dataset for each client $\mathcal{D}$; Loss function $\mathcal{L}$; Global model in the r-th round W_r; Encoder $En(\cdot)$; Classification head $W_{cls}(\cdot)$; Queue size q; Dynamic fusion hyperparameter γ.

for each communication round $r = 1, 2, \ldots, R$ **do**
  $m \leftarrow \max(P \times N, 1)$
  Randomly select a subset of m clients $\mathcal{N}_r \subseteq \mathcal{N}$
  Send global model $W_r = W_{cls}(En(\cdot))$ to $\mathcal{N}_r$
  for each selected client $k \in \mathcal{N}_r$ in parallel **do**
    # receive global model
    $W^k_{r,0} \leftarrow W_r$
    Initialize feature queue $\mathbf{U}$ with size Q
    for each local epoch $e = 1, 2, \ldots, E$ **do**
      for each batch $\mathcal{B} \in \mathcal{D}_k$ **do**
        $\mathbf{Q} \leftarrow En(\mathcal{B})$, $\mathbf{L} \leftarrow W_{cls}(\mathcal{Q})$
        $\mathcal{L}(W^k_{r,e-1}) \leftarrow \mathcal{L}_{\mathrm{CCL}}(\mathbf{Q}, \mathbf{L}, \mathbf{U}, \gamma)$
        ▷ According to Equations (30-32)
        $W^k_{r,e} \leftarrow W^k_{r,e-1} - \eta\nabla\mathcal{L}(W^k_{r,e-1})$
        Update feature queue $\mathbf{U}$
      end for
    end for
    Send trained local model $W^k_{r,E}$ to the server
  end for
  # global model aggregation
  $W_{r+1} \leftarrow \frac{1}{m}\sum_{k\in\mathcal{N}_r} W^k_{r,E}$
end for

IV. Experiments

A. Experimental Settings

1) Datasets and Models: We adopt two widely used RS image classification datasets to validate the performance of our RS-CCL, namely UC Merced Land Use (UC Merced) dataset [45] and NWPU [48] dataset.

The UC Merced dataset, released by the University of California, Merced in 2010, is a publicly available dataset frequently used for RS image classification. This dataset contains 21 types of land use scene images, such as agricultural areas, airports, forests, etc. Each category includes 100 images with a resolution of 256 × 256 pixels. Fig. 4 selects one example image for each class in the dataset. These images are all derived from the satellites of the United States Geological Survey (USGS). We divide this dataset into a training set and a test set with a ratio of 8:2, where the former is divided into a total of 42 clients and the latter is deployed on the central server.

The NWPU dataset is a benchmark dataset in the field of RS image scene classification, created and released by Northwestern Polytechnical University in China in 2017. The dataset encompasses 45 scene categories, including aircraft, airports, beaches, bridges, farmland, commercial districts, residential areas, and forests. Each category contains 700 images, totaling 31 500 RGB images with a pixel size of 256 × 256. Image spatial resolutions range from approximately 0.2–30 m, primarily sourced from platforms such as Google Earth. We

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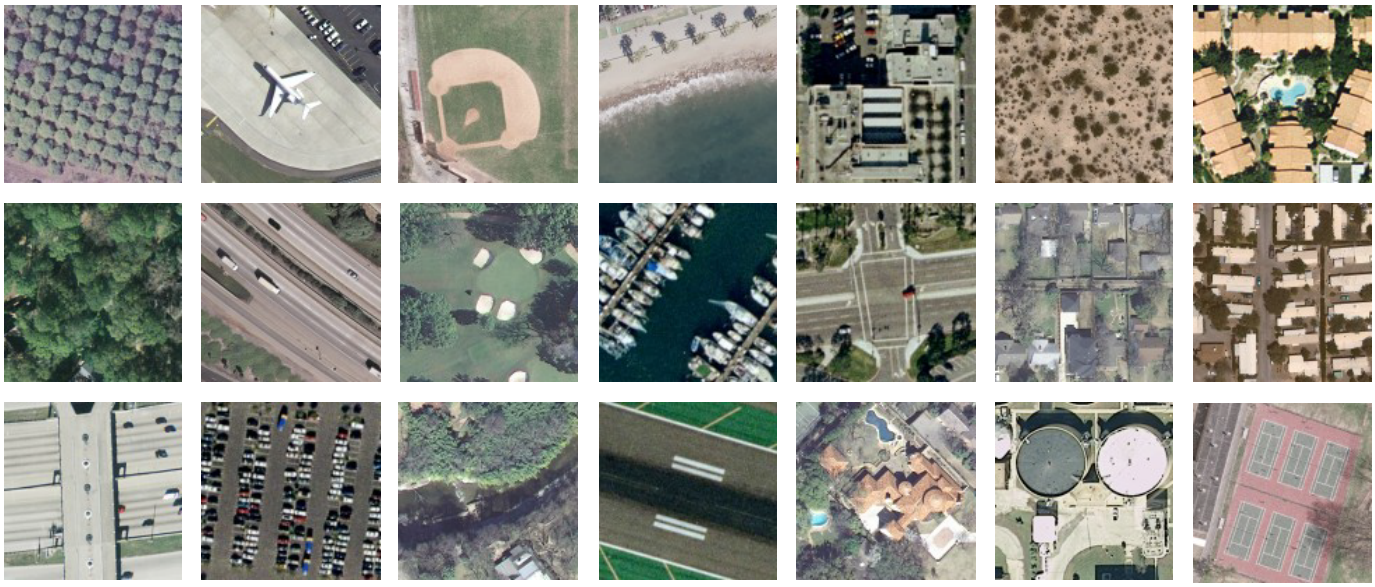


Fig. 4. Example images of UC Merced dataset containing 21 categories in total: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis court.

divide the NWPU dataset into a training set and a test set with a ratio of 8:2, where the former is divided into 100 clients, and the latter is deployed on the server.

As for models, we adopt three widely used backbones, namely ResNet18 [49], VGG16 [50], and MobileViT-S [51]. All compared methods, including our own, are evaluated on the two RS datasets using these three models.

2) Baselines and Implementation Details: We compare our proposed RS-CCL against several state-of-the-art FL methods, including FedAvg [19], FedProx [13], SCAFFOLD [14], MOON [41], FedPCL [43], FedCDG [44], FedProc [42], FedDecorr [46], FedDyn [52], and FedRCL [47]. FedAvg is the earliest FL algorithm. FedProx adds a regularization term to FedAvg. Scaffold corrects the "client drift" issue by introducing control variables. MOON applies contrastive learning at the model level. FedPCL starts from a unified pretrained model and promotes the personalization of each client model through contrastive learning. FedCDG combines FL with domain generalization, and uses contrastive learning to learn universal representations from data from multiple decentralized clients. FedProc achieves regularization through prototype contrastive learning. FedDecorr minimizes correlations between feature dimensions in model representations. FedDyn introduces a dynamic regularization term on the server side. FedRCL introduces a relaxed contrastive learning loss function.

We utilize parameter settings in FL to simulate real-world multisatellite scenarios. N clients represent N satellites. The client selection ratio P per round simulates the proportion of satellites capable of interacting with the central server satellite during each communication cycle. Data heterogeneity in FL mimics the real-world scenario where data collected by different satellites exhibits distinct distributions. We set the proportion of clients P to 0.24 and the client count N to 42 for experiments on the UC Merced dataset, and the proportion P to 0.1 and the client count N to 100 for experiments on the NWPU dataset. The total number of communication rounds R is set to 1,000. For local training of each client, we adopt SGD as the optimizer, and the initial value of the learning rate η is set to 0.05 with a decay of 0.998 times with each communication round. The number of local training epochs is set to 5, and the batch size is set to 10. For the data heterogeneity setting, we set up four settings, including three sets of non-IID experiments and a set of IID experiments. We divide the training set into different subsets by Dirichlet distribution to simulate the case of heterogeneous data in realistic scenarios, and the symmetric parameter α is set to $\{0.05, 0.1, 0.5\}$. A smaller α implies stronger data heterogeneity. The mean and standard deviation of the average top-1 accuracy over the last 100 rounds are computed from five experiments, each utilizing four different random seeds. We use the PyTorch framework, and the experiments are implemented on NVIDIA GeForce RTX 3090 GPUs.

For the hyperparameters of our proposed RS-CCL, the temperature τ is set to 0.05, γ is set to 0.05, and the queue size Q is set to 50. We also take an ablation study of different performances caused by different selections of Q in Section IV-C2, proving that $Q = 50$ performs satisfactorily in most of the experiments. Moreover, a set of hyperparameter sensitivity experiments of γ is presented in Section IV-C3, demonstrating that the performance of our method is not overly influenced by the selection of the hyperparameter γ.

B. Results and Analysis

The results on the UC Merced dataset are shown in Table I. It can be observed that the proposed method RS-CCL achieved optimal performance under all non-IID settings as well as in IID settings, with classification accuracy significantly surpassing other comparison methods. Specifically, in the highly heterogeneous scenario ($\alpha = 0.05$) with the ResNet18 model, RS-CCL reaches an accuracy of 87.17%,

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TABLE I
RESULTS ON UC MERCED DATASET

Model	Method	α = 0.05	α = 0.1	α = 0.5	IID
ResNet18	FedAvg [19]	79.75±2.2	80.16±2.0	81.50±1.8	82.74±1.0
	FedProx [13]	80.22±1.5	80.53±2.3	81.22±1.4	82.20±0.8
	SCAFFOLD [14]	79.97±2.4	80.99±1.9	85.23±1.1	85.09±1.1
	MOON [41]	80.75±1.7	81.66±2.0	86.07±1.7	86.11±0.5
	FedPCL [43]	80.06±2.0	81.25±1.5	85.21±1.5	85.35±1.0
	FedCDG [44]	80.53±2.3	81.65±2.0	85.97±1.0	85.86±0.7
	FedProc [42]	78.22±2.5	79.55±2.2	82.34±1.1	82.76±1.2
	FedDecorr [46]	82.08±1.5	82.24±1.7	86.15±1.5	86.22±0.6
	FedDyn [52]	82.19±2.1	83.33±1.9	86.27±1.7	86.13±0.9
	FedRCL [47]	83.50±2.3	84.29±1.9	86.97±0.6	88.00±0.9
	RS-CCL	**87.17±1.5**	**88.07±1.5**	**90.56±0.8**	**90.82±0.7**
VGG16	FedAvg [19]	78.90±2.3	79.40±2.1	80.80±1.9	82.10±1.1
	FedProx [13]	79.40±1.6	79.80±2.4	80.50±1.5	81.60±0.9
	SCAFFOLD [14]	78.80±2.5	80.10±2.0	84.50±1.2	84.30±1.2
	MOON [41]	79.60±1.8	80.70±2.1	85.30±1.8	85.40±0.6
	FedPCL [43]	79.24±1.9	80.11±1.5	85.17±1.2	85.20±1.1
	FedCDG [44]	79.97±1.6	80.54±1.8	85.36±1.1	85.25±0.9
	FedProc [42]	77.50±2.6	78.80±2.3	81.60±1.2	82.00±1.3
	FedDecorr [46]	81.30±1.6	81.50±1.6	85.40±1.1	85.50±0.7
	FedDyn [52]	81.40±2.2	82.50±2.0	85.60±1.8	85.40±1.0
	FedRCL [47]	82.70±2.4	83.50±2.0	86.20±1.7	87.20±1.3
	RS-CCL	**86.00±1.8**	**86.90±1.6**	**89.30±0.9**	**89.50±1.0**
MobileViT	FedAvg [19]	80.30±2.4	80.70±2.2	82.00±2.0	83.10±1.2
	FedProx [13]	80.80±1.7	81.20±2.5	81.90±1.6	82.60±1.0
	SCAFFOLD [14]	80.10±2.6	81.30±2.1	85.80±1.3	85.60±1.3
	MOON [41]	81.00±1.5	82.00±1.2	86.50±1.1	86.60±0.6
	FedPCL [43]	80.88±2.2	81.73±2.3	86.04±0.7	86.20±0.6
	FedCDG [44]	80.96±1.5	81.66±1.7	86.25±1.3	86.22±1.1
	FedProc [42]	79.00±2.7	80.30±2.4	83.00±1.3	83.30±1.4
	FedDecorr [46]	82.50±1.7	82.70±1.9	86.50±1.7	86.60±0.7
	FedDyn [52]	82.60±2.3	83.70±2.1	86.70±1.9	86.50±1.1
	FedRCL [47]	83.90±2.5	84.70±2.1	87.40±0.8	88.40±1.1
	RS-CCL	**87.50±2.1**	**88.40±1.6**	**90.90±1.3**	**91.10±0.8**

TABLE II
RESULTS ON NWPU DATASET

Model	Method	α = 0.05	α = 0.1	α = 0.5	IID
ResNet18	FedAvg [19]	78.34±1.7	80.95±0.5	82.78±0.4	83.53±0.4
	FedProx [13]	79.07±2.6	81.55±1.1	82.82±0.3	83.12±0.4
	SCAFFOLD [14]	80.01±2.9	81.09±2.1	83.87±0.9	84.24±0.6
	MOON [41]	81.95±1.8	82.88±2.0	84.04±1.5	84.35±0.8
	FedPCL [43]	80.54±1.3	81.79±1.5	83.89±1.5	84.14±0.7
	FedCDG [44]	82.00±2.7	82.84±2.5	84.17±1.2	84.50±1.2
	FedProc [42]	77.97±2.0	80.99±0.9	83.82±0.8	84.22±0.8
	FedDecorr [46]	80.00±1.7	81.93±1.5	84.23±0.6	84.35±0.7
	FedDyn [52]	80.26±2.5	82.08±1.9	84.60±1.6	84.32±0.7
	FedRCL [47]	82.44±2.9	83.26±2.5	85.83±1.3	86.05±0.5
	RS-CCL	**85.04±2.0**	**86.67±1.1**	**89.48±1.0**	**89.96±0.2**
VGG16	FedAvg [19]	77.85±2.1	80.42±1.3	82.13±0.8	83.07±0.6
	FedProx [13]	78.63±2.5	81.02±1.4	82.25±0.6	82.83±0.5
	SCAFFOLD [14]	79.18±2.8	80.65±2.1	83.14±1.1	83.76±0.7
	MOON [41]	80.53±1.8	82.03±2.2	83.57±1.3	84.02±0.8
	FedPCL [43]	79.93±2.2	81.29±2.0	83.23±1.5	83.22±1.1
	FedCDG [44]	80.69±2.0	81.99±1.1	83.59±1.3	84.20±1.1
	FedProc [42]	77.43±2.3	80.18±1.1	83.04±0.9	83.72±0.8
	FedDecorr [46]	79.34±1.9	81.52±1.7	83.92±0.8	84.13±0.6
	FedDyn [52]	79.62±2.7	81.83±2.1	84.13±1.6	84.05±0.9
	FedRCL [47]	81.53±2.9	82.83±2.4	85.16±1.2	85.64±0.7
	RS-CCL	**84.03±2.0**	**85.57±1.2**	**88.63±1.0**	**89.14±0.6**
MobileViT	FedAvg [19]	79.13±2.2	81.43±1.5	83.24±1.1	84.06±0.7
	FedProx [13]	79.82±2.6	82.03±1.6	83.36±0.9	83.94±0.6
	SCAFFOLD [14]	80.64±2.9	81.84±2.3	84.43±1.2	84.91±0.8
	MOON [41]	82.34±2.1	83.23±2.4	84.86±1.5	85.13±0.9
	FedPCL [43]	81.97±2.5	83.02±1.4	84.23±1.1	84.90±1.5
	FedCDG [44]	82.30±2.1	83.19±2.5	84.90±1.3	85.21±0.8
	FedProc [42]	78.83±2.5	81.53±1.2	84.23±1.1	84.85±0.8
	FedDecorr [46]	81.04±2.0	82.54±1.8	84.83±0.9	85.02±0.7
	FedDyn [52]	81.35±2.8	83.04±2.2	85.13±1.8	84.94±1.0
	FedRCL [47]	83.25±3.1	84.13±2.7	86.42±1.3	86.83±0.7
	RS-CCL	**86.04±2.3**	**87.52±1.5**	**90.13±1.2**	**90.57±1.4**

improving by approximately 3.67%–8.95% compared to other methods. In the more uniform distribution (α = 0.5) and IID settings, RS-CCL further increases accuracy to 90.56% and 90.82% with the ResNet18 model, indicating its good adaptability and generalization ability in relatively balanced data distribution scenarios. Notably, FedProc performs relatively poorly on this dataset, even falling below the classic FedAvg method, which we believe may be due to the challenges faced by prototype learning-based FL methods when adapting to high-resolution data. The experimental results on VGG16 and MobileViT-S models under different data heterogeneity settings are similar to those of ResNet18, which verifies the excellent generalization performance of our method. Overall, RS-CCL significantly enhances the robustness and convergence performance of models in FL by constraining the consistency of update directions.

From the results shown in Table II of the NWPU dataset, it can be seen that RS-CCL consistently outperforms existing methods across all data distribution settings, further validating its effectiveness and generalization capability. In the extreme non-IID case (α = 0.05) with the ResNet18 model, RS-CCL achieves an accuracy of 85.04%, surpassing the second-best FedRCL by approximately 2.6 percentage points. When it comes to the MobileViT-S model, this lead reached approximately 2.8%. Meanwhile, in both the α = 0.5 and IID settings, their accuracies reach 89.48% and 89.96% with the ResNet18 model, respectively. Similar to the UC Merced results, FedProc continues to perform poorly. These results demonstrate that RS-CCL maintains stable and excellent performance when handling different intensities and types of distribution shifts.

C. Ablation Study

1) Ablation Results of Each Component: According to the ablations in Section III-C, we also perform statistics on the previously mentioned metrics of our proposed RS-CCL to prove that the coupling learning strategy we narrated earlier is able to alleviate the phenomenon of dimensional collapse. We conduct experiments on the UC Merced dataset with the ResNet18 model, and report the results of k_{90}, k_{95}, and Effective Rank in the 1000th round, along with the average accuracy on the last 100 rounds. The integrated results under $\alpha \in \{0.05, 0.5\}$ and IID settings are shown in Table III. It can be observed that the proposed RS-CCL, based on our analysis, effectively expands the dimension of features spatially (as evidenced by improvements in k_{90}, k_{95}, and effective rank metrics). At the same time, the training accuracy is further improved.

2) Analysis of Different Queue Sizes: As mentioned above, the MoCo framework is usually open-ended with respect to queue length settings (usually an integer multiple of batch

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TABLE III
ABLATION RESULTS ON UC MERCED DATASET WITH RESNET18, AND "E. RANK" REFERS TO EFFECTIVE RANK

	α = 0.05				α = 0.5				IID			
	k_{90}	k_{95}	E. Rank	Acc.(%)	k_{90}	k_{95}	E. Rank	Acc.(%)	k_{90}	k_{95}	E. Rank	Acc.(%)
FedAvg ($\mathcal{L}_{\text{CE}}$)	5	8	20.8629	79.75±2.2	9	13	27.4489	81.50±1.8	10	14	35.7134	82.74±1.0
$\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{UCL}}$	4	6	20.1514	82.39±1.9	6	9	25.1542	86.76±1.3	7	10	30.4303	88.55±1.4
$\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{ICCL}}$	4	6	20.3548	83.51±1.0	6	7	21.5254	87.10±1.1	5	7	21.0274	89.27±0.3
$\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{UCL}} + \mathcal{L}_{\text{ICCL}}$	4	6	18.3134	83.07±1.7	5	6	19.6185	86.68±0.4	5	6	19.7743	89.20±0.9
$\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{SSMoCo}}$	6	10	27.0043	85.00±1.5	13	15	40.4484	88.55±1.2	13	15	45.6815	88.72±0.6
RS-CCL	**8**	**10**	**31.0952**	**87.17±1.5**	**13**	**16**	**44.3045**	**90.56±0.8**	**14**	**17**	**49.2120**	**90.82±0.7**

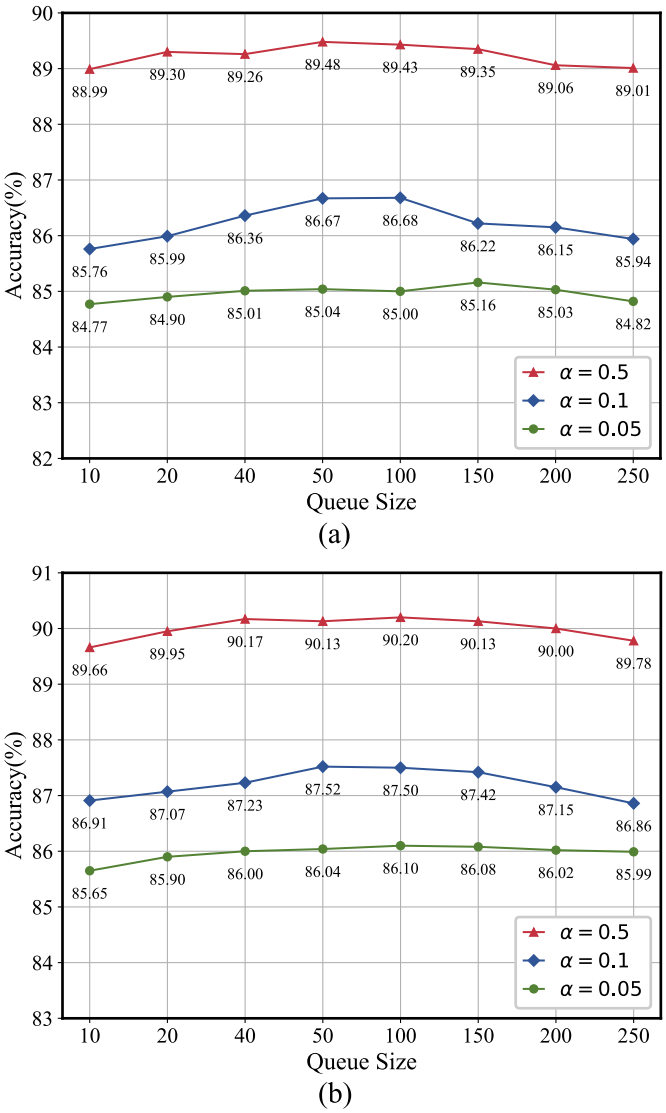


Fig. 5. Ablation results for different queue size Q on UC Merced dataset with (a) ResNet18 and (b) MobileViT-S.

size for ease in maintenance and updating). According to the batch size in our settings ($|\mathcal{B}| = 10$), we conduct experiments under different data heterogeneity in the value range of $\{10, 20, 40, 50, 100, 150, 200, 250\}$. We conduct related experiments on the NWPU dataset with ResNet18 and MobileViT-S models, and the results are shown in Fig. 5. It can be seen that $Q = 50$ achieves relatively good performance in all experimental settings. Furthermore, as the queue length increases significantly, the performance slightly declines, but it remains stable overall. On the whole, our RS-CCL is quite robust to different selections of queue length.

TABLE IV
ABLATION RESULTS (%) WITH DIFFERENT γ ON UC MERCED DATASET

γ	0	0.001	0.01	0.05	0.1	0.5	1
α = 0.5	88.48	89.64	89.97	**90.56**	90.38	90.24	89.31
α = 0.1	86.68	87.12	87.75	88.07	**88.12**	87.86	87.04
α = 0.05	85.89	86.59	87.00	**87.17**	86.95	86.36	86.11

3) Hyperparameter Sensitivity Analysis: γ is used to control the degree of emphasis placed on intraclass feature similarity within the loss function of RS-CCL. To investigate the impact of the hyperparameter γ on the model performance, we conducted a parameter sensitivity experiment on the UC Merced dataset with the ResNet18 model under different data heterogeneity settings. We select the values of γ from the set $[0, 0.001, 0.01, 0.05, 0.1, 0.5, 1]$, and the results are shown in Table IV. Analysis reveals that the value of γ has a significant effect on the model performance, showing a trend of increasing first and then decreasing overall. When $\alpha = 0.5$ and $\gamma = 0$, the model achieves the lowest 88.48% top-1 accuracy rate. As γ increases, the performance gradually improves, reaching a peak of 90.56% at $\gamma = 0.05$. However, when γ continues to increase to 1.0, the accuracy rate drops to 89.31%. This phenomenon indicates that introducing an appropriate γ constraint is beneficial for model optimization, but an excessively large γ value may impose overly strict limitations on the model, leading to a decline in performance. It is worth noting that when γ is within the range of 0.01–0.1, the model performance remains at a relatively high level (89.97%–90.56%), showing a gradual change. This demonstrates that the model is insensitive to variations in the γ value within this range and exhibits strong robustness. For the cases of $\alpha = 0.1$ and $\alpha = 0.05$, the experimental results are basically consistent with the trend of $\alpha = 0.5$. In a word, γ is an important but easily adjustable hyperparameter.

V. CONCLUSION

In this article, we present RS-CCL as a novel coupled learning rule in FL for RS image classification. After analyzing the impact of the feature dimension space on the model performance under data heterogeneity scenarios in FL, we formulate cross-entropy and feature similarity into a unified framework. Besides, a dynamic fusion mask matrix that changes according

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to the model input is introduced, while inhibiting the model from excessively drawing on interclass similarity to some extent. Extensive experiments across several RS classification datasets are conducted to validate the effectiveness and robustness of our proposed method, demonstrating significant improvements. In addition, we design ablation experiments to demonstrate the effectiveness of RS-CCL. These findings highlight the potential of RS-CCL to extend feature dimension and provide a powerful solution in improving network performance with complex RS images.

Future research will extend FL beyond single-modality image classification to multimodal RS data, such as optical-SAR or image–text representations, where coupled contrastive learning may facilitate cross-modal feature alignment in a federated setting. In addition, continual and incremental FL is highly relevant for realistic satellite systems, where new scenes, regions, or classes emerge over time. Exploring how coupled contrastive objectives can maintain feature diversity while mitigating catastrophic forgetting remains an important topic. These directions may further enhance the applicability of RS-CCL in large-scale and practical RS scenarios.

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